

Homework 2: Regression

Alex Czerkawski

Dataset choice

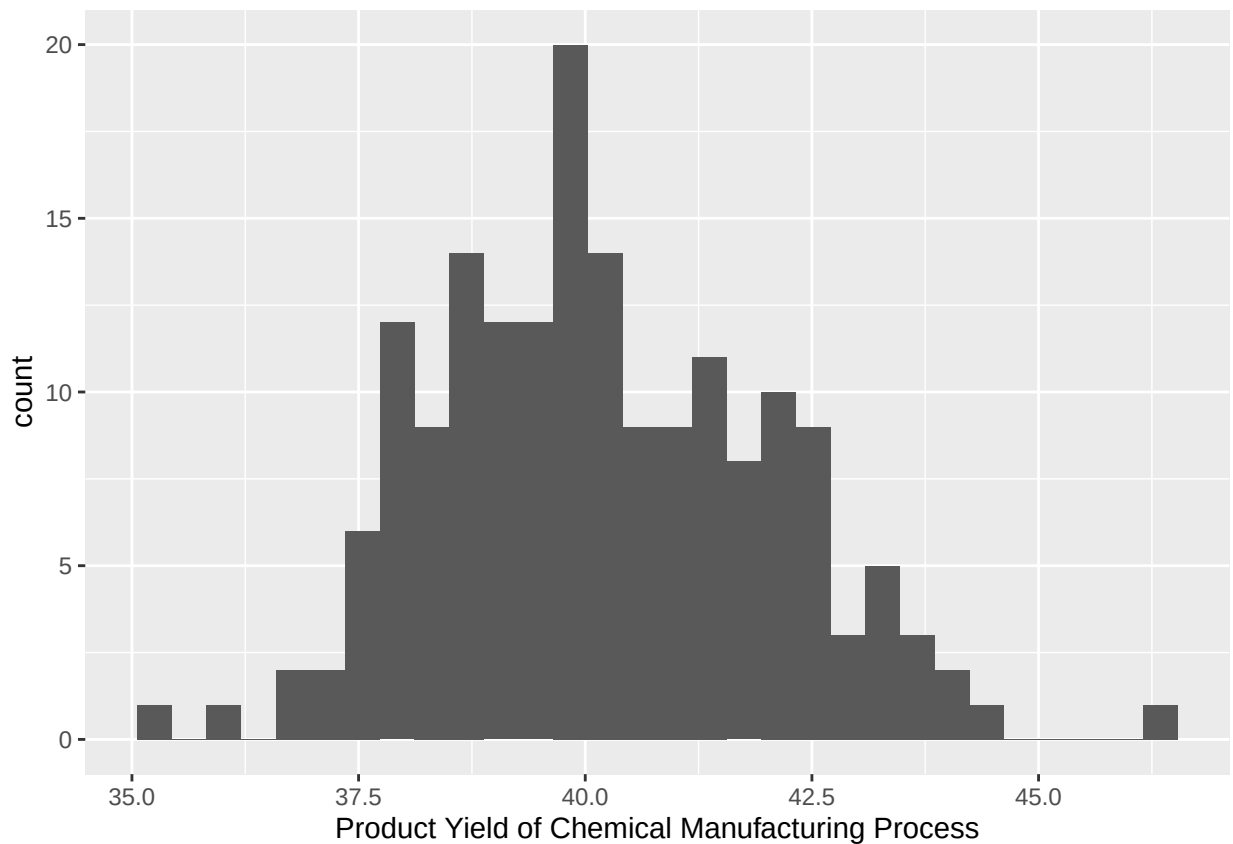
For this assignment, we will be using the Chemical Manufacturing Process dataset. This dataset was taken from Applied Predictive Models (APM) Chapter 1.4.

Identify an outcome variable to be predicted

- **Yield** (continuous) = Final product yield. A summary of this column is:

##	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
##	35.25	38.75	39.97	40.18	41.48	46.34

Below is a histogram of Yield



Preprocessing

The ChemicalManufacturingProcess data frame was pre-processed by removing all rows with missing values. Further pre-processing was applied when each model was fitted.

Linear Regression

```
## Linear Regression
##
## 152 samples
## 57 predictor
##
## Pre-processing: centered (57), scaled (57)
## Resampling: Cross-Validated (10 fold)
## Summary of sample sizes: 136, 137, 136, 137, 137, 136, ...
## Resampling results:
##
##      RMSE      Rsquared   MAE
## 3.587972 0.4697095 1.73205
##
## Tuning parameter 'intercept' was held constant at a value of TRUE
```

Lasso

```
## Elasticnet
##
## 152 samples
## 57 predictor
##
## Pre-processing: centered (57), scaled (57)
## Resampling: Cross-Validated (10 fold)
## Summary of sample sizes: 137, 138, 137, 136, 137, 137, ...
## Resampling results across tuning parameters:
##
## fraction    RMSE      Rsquared   MAE
## 0.01000000  1.704992  0.5680381  1.3983943
## 0.06210526  1.222923  0.6115414  1.0219197
## 0.11421053  1.172967  0.6113801  0.9810690
## 0.16631579  1.194311  0.5971413  0.9817072
## 0.21842105  1.209478  0.5970558  0.9858829
## 0.27052632  1.214108  0.5897723  1.0002780
## 0.32263158  1.377678  0.5440636  1.0677206
## 0.37473684  1.451118  0.5306503  1.0997577
## 0.42684211  1.560979  0.5179691  1.1402027
## 0.47894737  1.616090  0.5062786  1.1707529
## 0.53105263  1.558422  0.4973988  1.1695587
## 0.58315789  1.497879  0.4921429  1.1664369
## 0.63526316  1.499759  0.4855419  1.1783014
## 0.68736842  1.445487  0.5028335  1.1664311
## 0.73947368  1.605678  0.4626292  1.2259372
## 0.79157895  1.834278  0.4504471  1.2947070
## 0.84368421  2.072610  0.4444673  1.3646277
```

```
## 0.89578947 2.351265 0.4393362 1.4438196
## 0.94789474 2.612312 0.4353402 1.5169682
## 1.00000000 2.863655 0.4332464 1.5851985
##
## Tuning parameter 'lambda' was held constant at a value of 0
## RMSE was used to select the optimal model using the smallest value.
## The final values used for the model were fraction = 0.1142105 and lambda = 0.
```

Ridge Regression

```
## Ridge Regression
##
## 152 samples
## 57 predictor
##
## Pre-processing: centered (57), scaled (57)
## Resampling: Cross-Validated (10 fold)
## Summary of sample sizes: 137, 138, 137, 136, 137, 137, ...
## Resampling results across tuning parameters:
##
##   lambda      RMSE      Rsquared    MAE
## 0.000000000 2.863655 0.4332464 1.585198
## 0.005263158 1.438668 0.5219237 1.117959
## 0.010526316 1.278895 0.5630963 1.052056
## 0.015789474 1.223716 0.5941211 1.018480
## 0.021052632 1.212518 0.6010229 1.008866
## 0.026315789 1.220258 0.5962425 1.014270
## 0.031578947 1.234314 0.5902108 1.018740
## 0.036842105 1.249612 0.5854397 1.022267
## 0.042105263 1.264329 0.5819871 1.025035
## 0.047368421 1.277874 0.5795227 1.027129
## 0.052631579 1.290117 0.5777541 1.028688
## 0.057894737 1.301104 0.5764738 1.029819
## 0.063157895 1.310940 0.5755408 1.030770
## 0.068421053 1.319747 0.5748593 1.031661
## 0.073684211 1.327644 0.5743635 1.032594
## 0.078947368 1.334739 0.5740069 1.033573
## 0.084210526 1.341130 0.5737564 1.034411
## 0.089473684 1.346901 0.5735877 1.035082
## 0.094736842 1.352126 0.5734829 1.035774
## 0.100000000 1.356870 0.5734284 1.036499
##
## RMSE was used to select the optimal model using the smallest value.
## The final value used for the model was lambda = 0.02105263.
```

Support Vector Machines (SVM)

```
## Support Vector Machines with Radial Basis Function Kernel
##
## 152 samples
## 57 predictor
##
```

```

## Pre-processing: centered (57), scaled (57)
## Resampling: Cross-Validated (10 fold)
## Summary of sample sizes: 137, 138, 137, 136, 137, 137, ...
## Resampling results across tuning parameters:
##
##      C      RMSE      Rsquared    MAE
##      0.25  1.421628  0.5215577  1.1505328
##      0.50  1.309358  0.5548664  1.0749098
##      1.00  1.185263  0.6272764  0.9547212
##      2.00  1.104157  0.6714562  0.8877469
##      4.00  1.076491  0.6890841  0.8558714
##      8.00  1.065618  0.6980059  0.8496085
##     16.00  1.062977  0.6992103  0.8473979
##     32.00  1.062977  0.6992103  0.8473979
##
## Tuning parameter 'sigma' was held constant at a value of 0.01291609
## RMSE was used to select the optimal model using the smallest value.
## The final values used for the model were sigma = 0.01291609 and C = 16.

```

Neural Nets

```

## Neural Network
##
## 152 samples
## 57 predictor
##
## Pre-processing: centered (57), scaled (57)
## Resampling: Cross-Validated (10 fold)
## Summary of sample sizes: 137, 138, 137, 136, 137, 137, ...
## Resampling results across tuning parameters:
##
##      decay  size  RMSE      Rsquared    MAE
##      5      1    1.632517  0.4026305  1.283935
##      5      2    2.064916  0.2690571  1.510084
##      5      3    2.489334  0.2813639  1.983615
##     20      1    3.247797  0.5001004  2.769568
##     20      2    2.423419  0.5104071  1.941508
##     20      3    2.276555  0.4559187  1.695527
##     80      1    9.374462  0.5613816  9.194590
##     80      2    6.972544  0.5505581  6.747530
##     80      3    5.619109  0.5798931  5.366477
##
## RMSE was used to select the optimal model using the smallest value.
## The final values used for the model were size = 1 and decay = 5.

```

Multivariate Adaptive Regression Splines (MARS)

```

## Multivariate Adaptive Regression Spline
##
## 152 samples
## 57 predictor
##

```

```

## No pre-processing
## Resampling: Cross-Validated (10 fold)
## Summary of sample sizes: 137, 138, 137, 136, 137, 137, ...
## Resampling results across tuning parameters:
##
##   degree  nprune  RMSE      Rsquared  MAE
##   1       20      1.215720  0.6134395  0.9913776
##   1       30      1.215720  0.6134395  0.9913776
##   1       40      1.215720  0.6134395  0.9913776
##   1       50      1.215720  0.6134395  0.9913776
##   2       20      1.346010  0.5426903  1.0594952
##   2       30      1.389933  0.5262297  1.0927374
##   2       40      1.389933  0.5262297  1.0927374
##   2       50      1.389933  0.5262297  1.0927374
##
## RMSE was used to select the optimal model using the smallest value.
## The final values used for the model were nprune = 20 and degree = 1.

```

K-nearest Neighbors (KNN)

```

## k-Nearest Neighbors
##
## 152 samples
## 57 predictor
##
## Pre-processing: centered (57), scaled (57)
## Resampling: Cross-Validated (10 fold)
## Summary of sample sizes: 137, 138, 137, 136, 137, 137, ...
## Resampling results across tuning parameters:
##
##   k  RMSE      Rsquared  MAE
##   5  1.277457  0.5793753  1.039724
##   7  1.295666  0.5741845  1.049574
##   9  1.289960  0.5715154  1.057556
##  11  1.342480  0.5360619  1.106669
##  13  1.367563  0.5208765  1.123378
##  15  1.374703  0.5140158  1.130329
##  17  1.408501  0.4827092  1.153826
##  19  1.406533  0.4916467  1.152317
##  21  1.408395  0.5006288  1.146420
##  23  1.421859  0.4975980  1.162114
##  25  1.440681  0.4864208  1.178190
##  27  1.449021  0.4871062  1.187704
##  29  1.465612  0.4724545  1.206630
##  31  1.476393  0.4667232  1.219566
##  33  1.480504  0.4725088  1.224523
##
## RMSE was used to select the optimal model using the smallest value.
## The final value used for the model was k = 5.

```

Regression tree

```
## CART
```

```
##
## 152 samples
## 57 predictor
##
## No pre-processing
## Resampling: Cross-Validated (10 fold)
## Summary of sample sizes: 137, 138, 137, 136, 137, 137, ...
## Resampling results across tuning parameters:
##
##   cp          RMSE      Rsquared    MAE
##   0.00000000  1.477057  0.4534626  1.109530
##   0.02577868  1.444794  0.4601939  1.114744
##   0.05155735  1.416514  0.4694798  1.133163
##   0.07733603  1.425702  0.4816324  1.140512
##   0.10311471  1.570508  0.3732888  1.265982
##   0.12889338  1.531968  0.3774744  1.237088
##   0.15467206  1.553184  0.3611384  1.269577
##   0.18045074  1.553184  0.3611384  1.269577
##   0.20622941  1.553184  0.3611384  1.269577
##   0.23200809  1.553184  0.3611384  1.269577
##   0.25778676  1.553184  0.3611384  1.269577
##   0.28356544  1.553184  0.3611384  1.269577
##   0.30934412  1.553184  0.3611384  1.269577
##   0.33512279  1.553184  0.3611384  1.269577
##   0.36090147  1.720758  0.2964437  1.411231
##
## RMSE was used to select the optimal model using the smallest value.
## The final value used for the model was cp = 0.05155735.
```

A bagged regression tree model was also fitted, resulting in a reduction in cross-validated RMSE.

```
## Bagged CART
##
## 152 samples
## 57 predictor
##
## No pre-processing
## Resampling: Cross-Validated (10 fold)
## Summary of sample sizes: 137, 138, 137, 136, 137, 137, ...
## Resampling results:
##
##   RMSE      Rsquared    MAE
##   1.148229  0.6485871  0.8968348
```

Random forests

```
## Random Forest
##
## 152 samples
## 57 predictor
##
## No pre-processing
## Resampling: Cross-Validated (10 fold)
```

```
## Summary of sample sizes: 137, 138, 137, 136, 137, 137, ...
## Resampling results across tuning parameters:
##
##   mtry  RMSE      Rsquared  MAE
##   10    1.0432347  0.7406963  0.8444729
##   15    1.0270728  0.7424761  0.8181692
##   20    1.0295272  0.7409280  0.8073260
##   25    0.9970958  0.7519230  0.7805817
##   30    1.0355982  0.7293404  0.8028158
##   36    1.0271672  0.7319162  0.8055174
##   41    1.0041544  0.7463935  0.7672223
##   46    1.0221329  0.7331611  0.7973225
##   51    1.0167054  0.7355987  0.7967598
##   57    1.0279694  0.7250956  0.8089502
##
## RMSE was used to select the optimal model using the smallest value.
## The final value used for the model was mtry = 25.
```

Model comparison

Below are the models listed in order of cross-validated RMSE from largest to smallest.

- Linear Regression (2.77)
- Neural Network (1.63)
- Regression Tree (1.42)
- KNN (1.28)
- MARS (1.22)
- Ridge Regression (1.21)
- Lasso (1.17)
- Bagged Regression Tree (1.15)
- SVM (1.06)
- Random Forest (1.00)

Below are the models listed in order of complexity from least to most complex.

- Linear Regression
- Regression Tree
- Ridge Regression
- Lasso
- KNN
- Neural Network
- MARS
- SVM
- Bagged Regression Tree
- Random Forest

Recommendation

As seen above, Random Forest produced the best cross-validated RMSE, though it's also the most complex model. The RMSE standard deviation for the Random Forest model was 0.26. If all models within the mean RMSE + RMSESD meet the selection criteria, then I would choose the simplest model with a cross-validated RMSE below 1.26. Therefore I would select either the Ridge Regression or LASSO model, as they are much more interpretable with a similar level of performance as the more complex models.